initial question of how animals have such low entropy. But if we follow the food chain, we ultimately come upon animals (like me) that eat only plants. How do plants and their products of fruits and vegetables maintain low entropy? Through photosynthesis, plants use sunlight to separate ambient carbon dioxide into oxygen, which is given back to the environment, and carbon, which the plants use to grow and flourish. So we can trace the low-entropy, nonanimal sources of energy to the sun.

This pushes the question of explaining low entropy another step further back: where did our highly ordered sun come from? The sun formed about 5 billion years ago from an initially diffuse cloud of gas that began to swirl and clump under the mutual gravitational attraction of all its constituents. As the gas cloud got denser, the gravitational pull of one part on another got stronger, causing the cloud to collapse further in on itself. And as gravity squeezed the cloud tighter, it got hotter. Ultimately, it got hot enough to ignite nuclear processes that generated enough outwardflowing radiation to stem further gravitational contraction of the gas. A hot, stable, brightly burning star was born.

So where did the diffuse cloud of gas come from? It likely formed from the remains of older stars that reached the end of their lives, went supernova, and spewed their contents out into space. Where did the diffuse gas responsible for these early stars come from? We believe that the gas was formed in the aftermath of the big bang. Our most refined theories of the origin of the universe—our most refined *cosmological* theories—tell us that by the time the universe was a couple of minutes old, it was filled with a *nearly uniform hot gas* composed of roughly 75 percent hydrogen, 23 percent helium, and small amounts of deuterium and lithium. The essential point is that this gas filling the universe had extraordinarily *low* entropy. The big bang started the universe off in a state of low entropy, and that state appears to be the source of the order we currently see. In other words, *the current order is a cosmological relic*. Let’s discuss this important realization in a little more detail.

Entropy and Gravity

Because theory and observation show that within a few minutes after the big bang, primordial gas was uniformly spread throughout the young